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Stochastic Finite Element Methods

#5: Stochastic Collocation Methods

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Introduction

- » Stochastic collocation methods are sampling based too, i.e. a deterministic solver will be called at certain points in the random space.
- » In contrast to MC based methods the systems response is approximated by a polynomial fit in the multi-dimensional stochastic space (response-surface).
- » Instead of generating random-numbers, the deterministic solution is computed on defined grids.
- » Computational effort and accuracy depends on the choice of the collocation points.

Polynomial Interpolation of the Response

Definition of the polynomial space:

$$\mathcal{P}_p(\Gamma^M) = \bigotimes_{i=1}^M \mathcal{P}_{p_i}(\Gamma_i), \quad \mathcal{P}_p(\Gamma^M) \subset \mathcal{L}_p^2(\Gamma^M)$$

where p is the order of the polynomials, M the dimension.

Subsidiary condition for p_i

$$\mathcal{P}_{p_i}(\Gamma_i) = \{v : \Gamma_1 \rightarrow \mathbb{R} : v \in \text{span}\{\phi_m(Y_i)\}_{m=0}^{p_i}\}, \quad i = 1, \dots, M$$

where Y_i^k is a one-dimensional point cloud of dimension $i \in M$ at level k .

$\{\phi_m(Y_i)\}$ define orthogonal polynomials which follow the condition:

$$\int_{\Gamma_i} \rho_i(Y_i) \phi_m(Y_i) \phi_n(Y_i) dY_i = h_m^2 \delta_{mn}, \quad h_m^2 = \int_{\Gamma_i} \rho_i \phi_m^2 dY_i$$

Polynomial Interpolation of the Response

The number of polynomials $\dim(\mathcal{P}(\Gamma^M)) = \prod_{i=1}^M (p_i + 1)$ increases with the order p_i of the polynomials and the stochastic dimension M (curse of dimensions).

Need for a proper choice of the collocation points, e.g. sparse grids.

Lagrange Polynomials

For N support points Y_i the Lagrange polynomials L_i out of the polynomial subspace $\mathcal{V}_{\mathcal{I}} \subset \mathcal{P}_N$ satisfy

$$L_i(Y) \in \mathcal{V}_{\mathcal{I}}, \quad L_i(Y_j) = \delta_{ij}, \quad 1 \leq i, j, \leq N$$

The coefficients are usually computed from divided difference formulas (Newton-scheme or Neville-Aitken scheme). Goal is to find an interpolation of kind

$$L^{\text{new}}(f(Y)) = a_0 + a_1 L_1(Y) + \cdots + a_{M-1} L_{M-1}(Y)$$

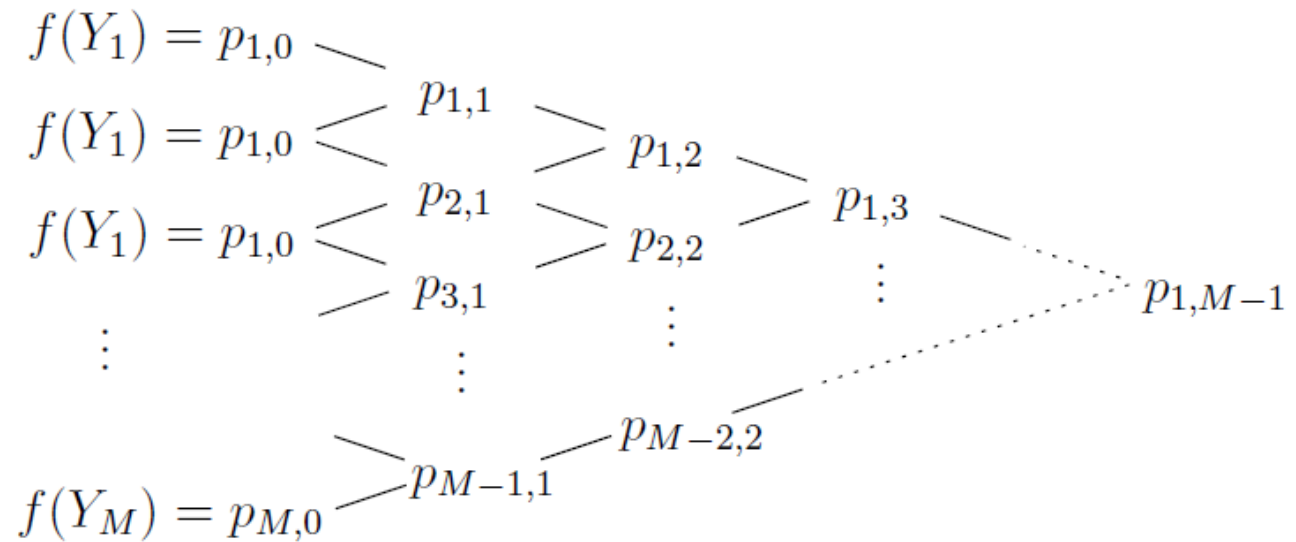
For one-dimensional support points $Y_1, \dots, Y_N$ one defines first the Newton polynomials [Quarteroni, Sacco and Saleri, 2007]

$$q_0(Y) = 1, \quad q_1(Y) = (Y - Y_1), \quad q_2(Y) = (Y - Y_1)(Y - Y_2) \cdots$$

$$Y_N(Y) = (Y - Y_1) \cdots (Y - Y_N)$$

Lagrange Polynomials

Aitken-Neville-Scheme



needs for the computation of only one new coefficient per polynomial

$$p_{1,k}(Y) - p_{1,k-1}(Y) = a_{1,k}q_k(Y) = a_{1,k}(Y - Y_1) \cdots (Y - Y_k)$$

Computation of the Lagrange-basis out of the Newton-basis (Horner-Scheme):

$$L^{\text{new}}(Y) = a_0 + a_1(Y - Y_1) + \cdots + a_{M-1}(Y - Y_1)(Y - Y_2) \cdots (Y - Y_{M-1})$$

Full Tensor Product

Goal: Find an Interpolant for a function in the M – dimensional hyperspace,
e.g. $f : [-1, 1]^M \rightarrow \mathbb{R}$ defined by the tensor product

$$\mathcal{P}_p^M(\Gamma^M) = \bigotimes_{i=1}^M \mathcal{P}_{p_i}(\Gamma_i), \quad \mathcal{P}_p(\Gamma^M) \subset \mathcal{L}_p^2(\Gamma^M)$$

Dimension of the problem: $\dim(\mathcal{P}_p(\Gamma_M)) = (1 + p)^M$, p : polynomial order

In one-dimensions: Choice for a set Θ with m_i support points Y_i
 $\rightarrow \Theta_1 = (Y_1, \dots, Y_{m_1}) \subset [-1, 1]$.

One defines a one-dimensional interpolant:

$$\mathcal{U}(f) = \sum_{k=1}^{m_1} L_{k_1} \cdot f(Y_k)$$

Full Tensor Product

For dimensions $M > 1$ one defines a tensor product space as

$$\Theta_M = (\Theta_m \otimes \cdots \otimes \Theta_m) \subset [-1, 1]^M$$

The multi-dimensional interpolant reads

$$\mathcal{I}(f) = (\mathcal{U}_1 \otimes \cdots \otimes \mathcal{U}_M)(f) = \sum_{k_1=1}^m \cdots \sum_{k_M=1}^m f(Y_{k_1}, \dots, Y_{k_M}) \cdot (L_{k_1} \otimes \cdots \otimes L_{k_M})$$

Computation of the expectation value, for example

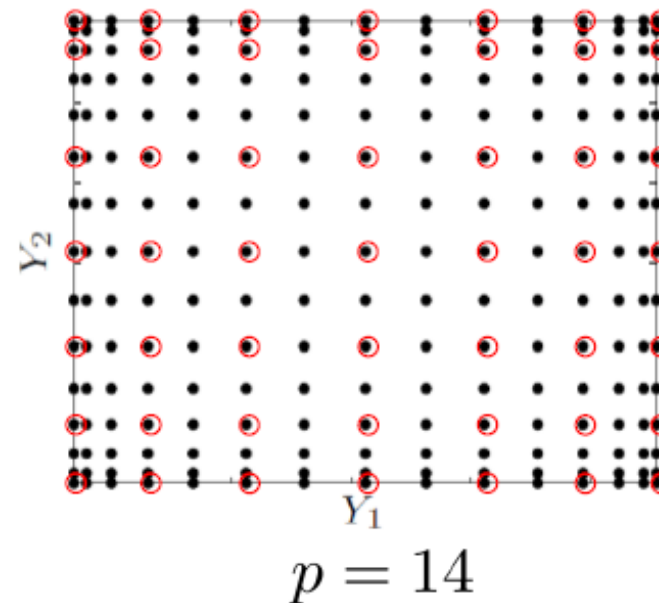
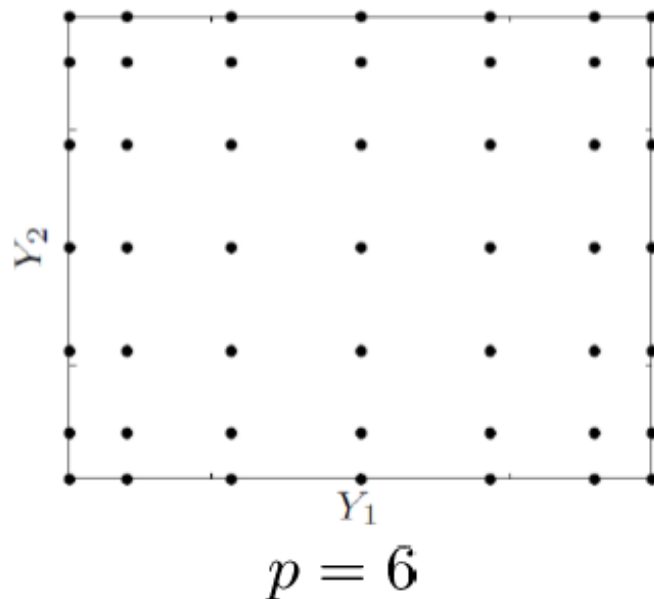
$$\mathbb{E}\{\mathcal{I}(f)\} = \sum_{k_1=1}^m \cdots \sum_{k_M=1}^m f(Y_{k_1}, \dots, Y_{k_M}) \int_{\Gamma} (L_{k_1} \otimes \cdots \otimes L_{k_M}) f_Y dY$$

Disadvantage: exponentially growth of computational effort with increasing dimension.

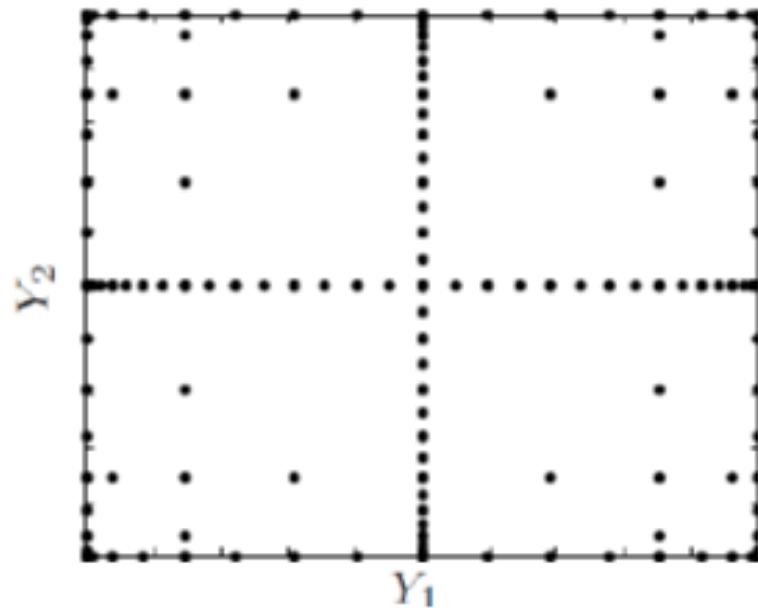
Full Tensor Product

Choice of the grid-points, e.g. roots of Chebyshev polynomials etc.

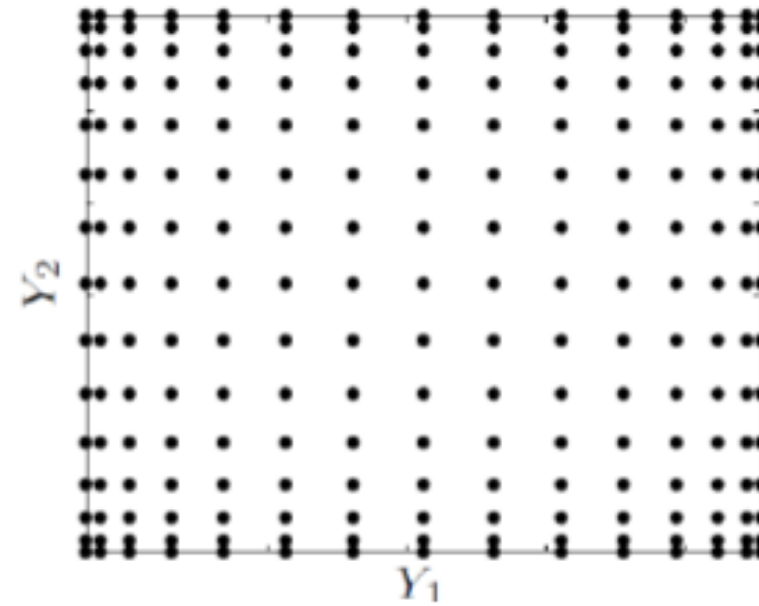
Hierarchical grid structures, here Clenshaw-Curtis grid



Sparse Grids



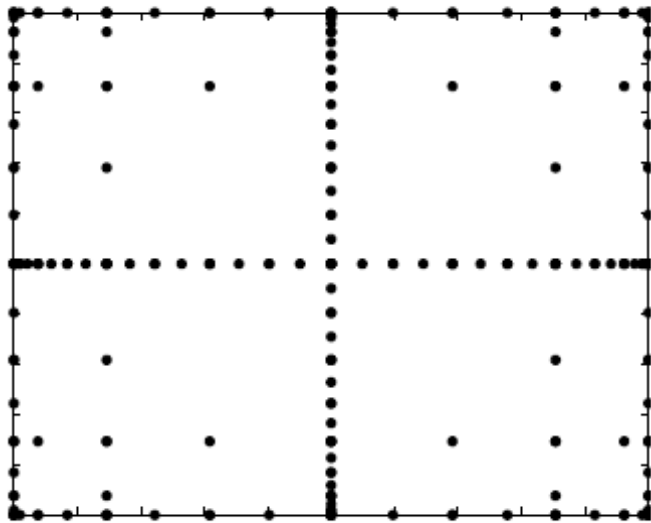
Sparse



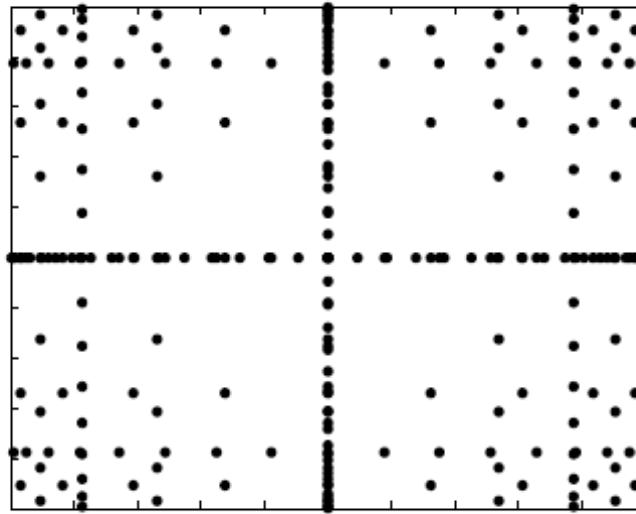
Full

Sparse Grids

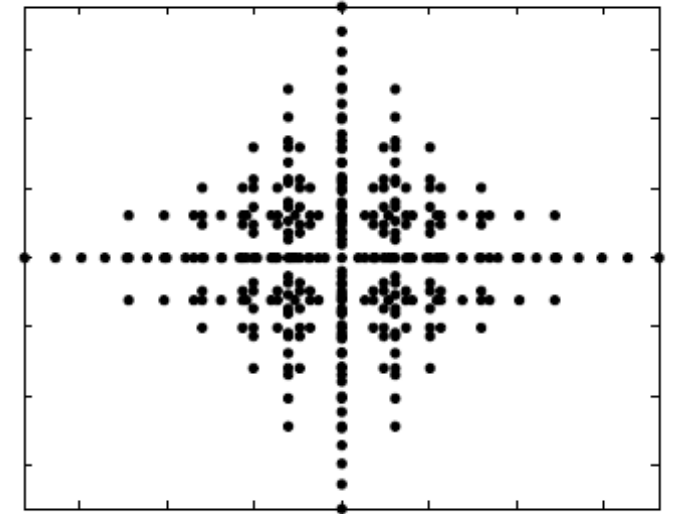
Choice of sparse grids, e.g.



Clenshaw-Curtis grid



Gauss grid



Gauss-Hermite grid

Smolyak Algorithm

Goal: less collocation points in comparison to full tensor product.

Polynomial space:

$$\mathcal{P}_k^M(\Gamma^M) = \mathcal{P}_p(\Gamma^M) = \bigotimes_{i=1}^M \mathcal{P}_{p_i}(\Gamma_i), \quad q - M + 1 \leq p \leq q$$

p : polynomial order; $k = p + 1$: so called level; $q = k + M$: abbreviation;
 M : stochastic dimension.

Interpolant [Xiu and Hesthaven 2005]:

$$\begin{aligned} \mathcal{I}(u) &= \sum_{k+1 \leq |i| \leq k+M} \eta(|i|) \cdot (\mathcal{U}^{i_1} \otimes \dots \otimes \mathcal{U}^{i_M}) \\ &= \sum_{k+1 \leq |i| \leq k+M} \eta(|i|) \cdot \sum_{k_1=1}^{m_{i_1}} \dots \sum_{k_M=1}^{m_{i_M}} f(Y_{k_1}^{i_1}, \dots, Y_{k_M}^{i_M}) \dots (L_{k_1}^{i_1} \otimes \dots \otimes L_{k_M}^{i_M}) \end{aligned}$$

with: $\eta(|i|) = (-1)^{q-|i|} \cdot \binom{N-1}{q-|i|}$ where: $|i| = i_1 + \dots + i_M$

Smolyak Algorithm

Statistical evaluation:

$$\mathbb{E}\{u\} = \sum_{q-M+1 \leq |i| \leq q} \eta(|i|) \cdot \sum_{k=1}^{m_{i_1}} \cdots \sum_{k=M}^{m_{i_M}} u(Y_{k_1}^{i_1}, \dots, Y_{k_M}^{i_M}) \cdot \int_{\Gamma} (L_{k_1}^{i_1} \otimes \cdots \otimes L_{k_M}^{i_M}) f_Y d\mathbf{Y}$$

$$\text{Var}\{u\} = \sum_{q-M+1 \leq |i| \leq q} \eta(|i|) \cdot \sum_{k=1}^{m_{i_1}} \cdots \sum_{k=M}^{m_{i_M}} [u(Y_{k_1}^{i_1}, \dots, Y_{k_M}^{i_M}) - \mathbb{E}\{u\}]^2 \cdot \int_{\Gamma} (L_{k_1}^{i_1} \otimes \cdots \otimes L_{k_M}^{i_M}) f_Y d\mathbf{Y}$$

As the stochastic dimension has not been changed, the evaluation of the high-dimensional integral may be costly, e.g. Monte Carlo Method needed.

Collocation Grid

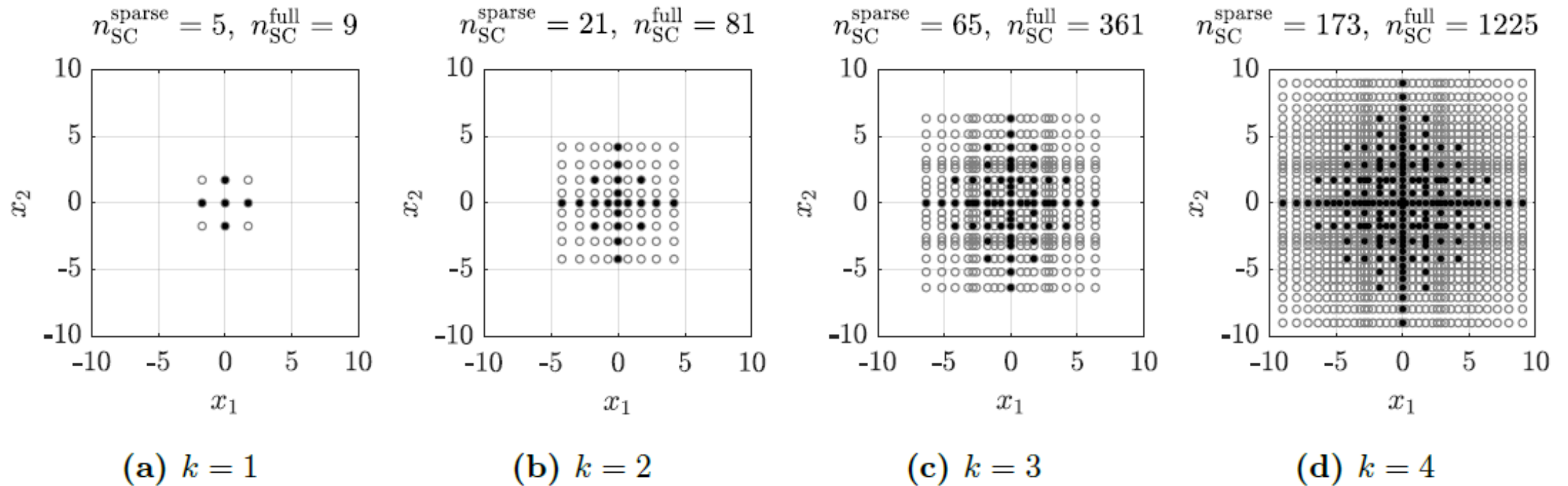
Choice of the grid-points, e.g. roots of Chebyshev polynomials etc.

Also Gaussian or Gaussian-based grids favorable for propagation of random fields; defined on $[-\infty, +\infty]$

Sparse grid	nested	symmetric	positive
Gauss-Hermite	✗	✓	✓
Kronrod-Patterson	✓	✓	✗
Gauss-Leja	✓	✗	✗

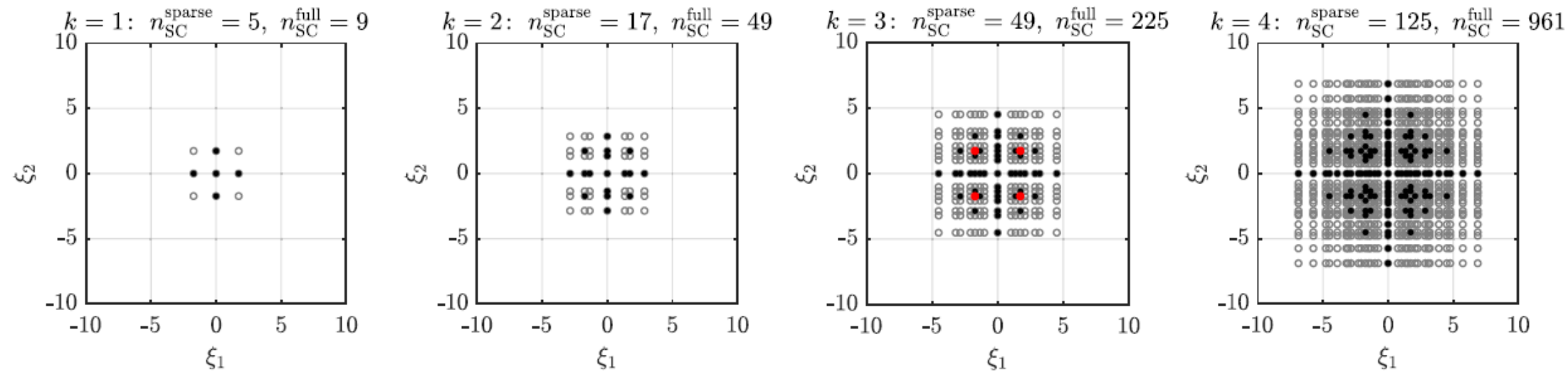
K-P Grid perhaps is the most robust for high-dimensions and non-linear problems (Dannert et al. 2022)

Collocation Grid



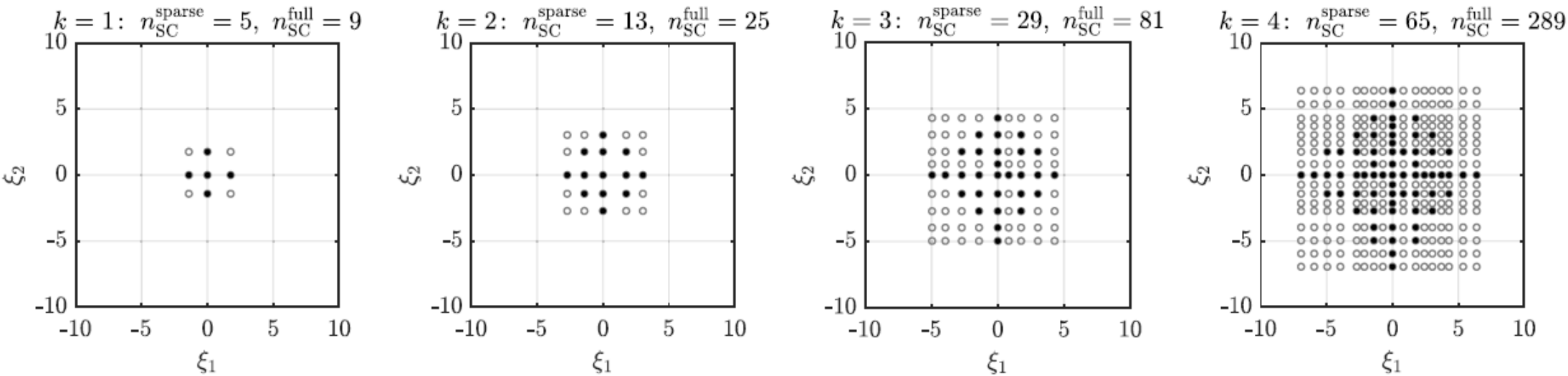
K-P grid (black dots) in comparison to full grid (circles) with increasing Smolyak level k
(Dannert et al. 2022)

Collocation Grid



Gauß-Hermit grid with increasing Smolyak level k

Collocation Grid



Gauß-Leja grids with increasing Smolyak level k

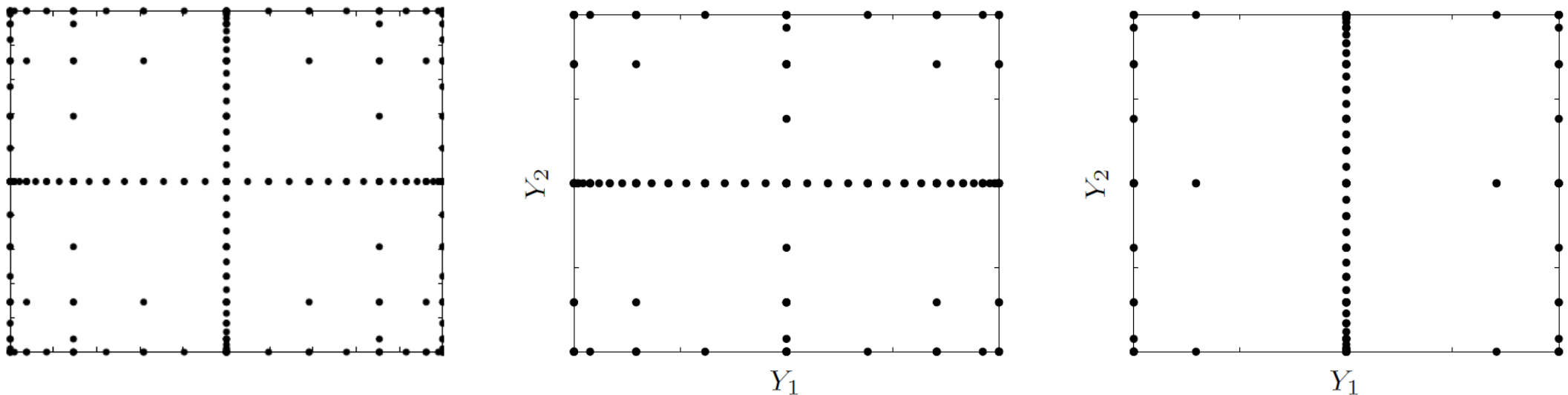
Sparse Grids

Sparse grid performance in comparison with full tensor grid:

Dim. M	Level k	Clenshaw-Curtis			Gauss-Hermite		
		$\#\mathcal{H}$	$\#\mathcal{H}_s$	$\#\mathcal{H}/\#\mathcal{H}_s$	$\#\mathcal{H}$	$\#\mathcal{H}_s$	$\#\mathcal{H}/\#\mathcal{H}_s$
2	1	7	5	1.4	7	5	1.4
	2	25	13	1.9231	25	17	1.4706
	3	67	29	2.3103	67	49	1.3673
	4	161	65	2.4769	161	125	1.288
3	1	10	7	1.4286	10	7	1.4286
	2	52	25	2.08	52	31	1.6774
	3	195	69	2.8261	195	111	1.7568
	4	609	177	3.4407	609	351	1.735
5	1	16	11	1.4545	16	11	1.4545
	2	131	61	2.1475	131	71	1.8451
	3	746	241	3.0954	746	351	2.1254
	4	3376	801	4.2147	3376	1471	2.295

Anisotropic Smolyak Algorithm

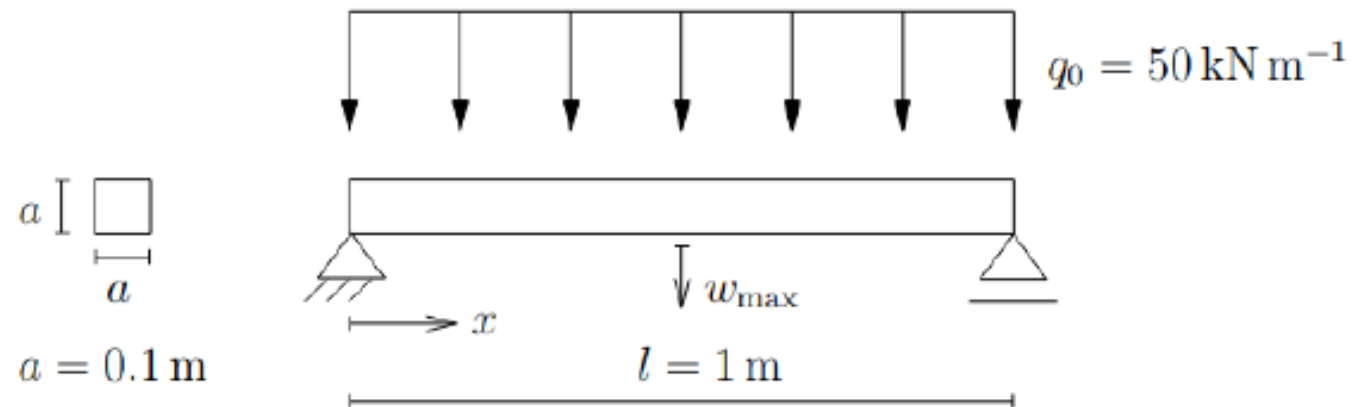
Idea: Reduced polynomial order in the dimensions of less or smoother importance for the systems response, e.g. [Babuska et al. 2007].



Cleshaw-Curtis Grids for different levels of anisotropy.

Numerical Example

Linear elastic problem with Young's modulus as random field

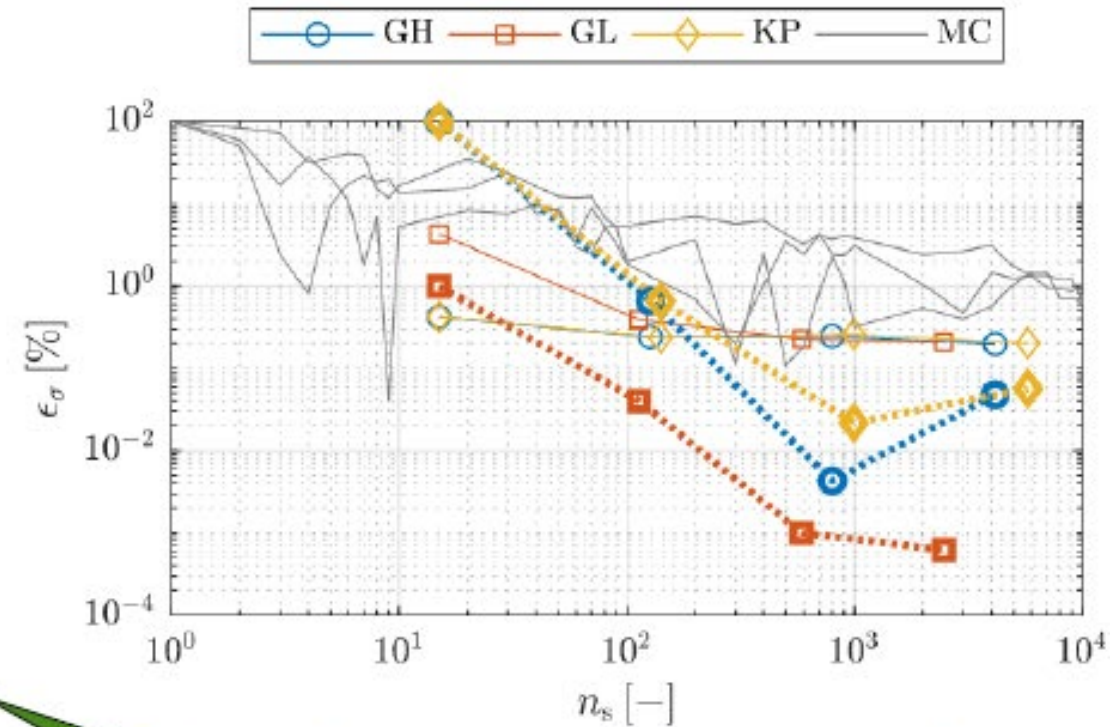
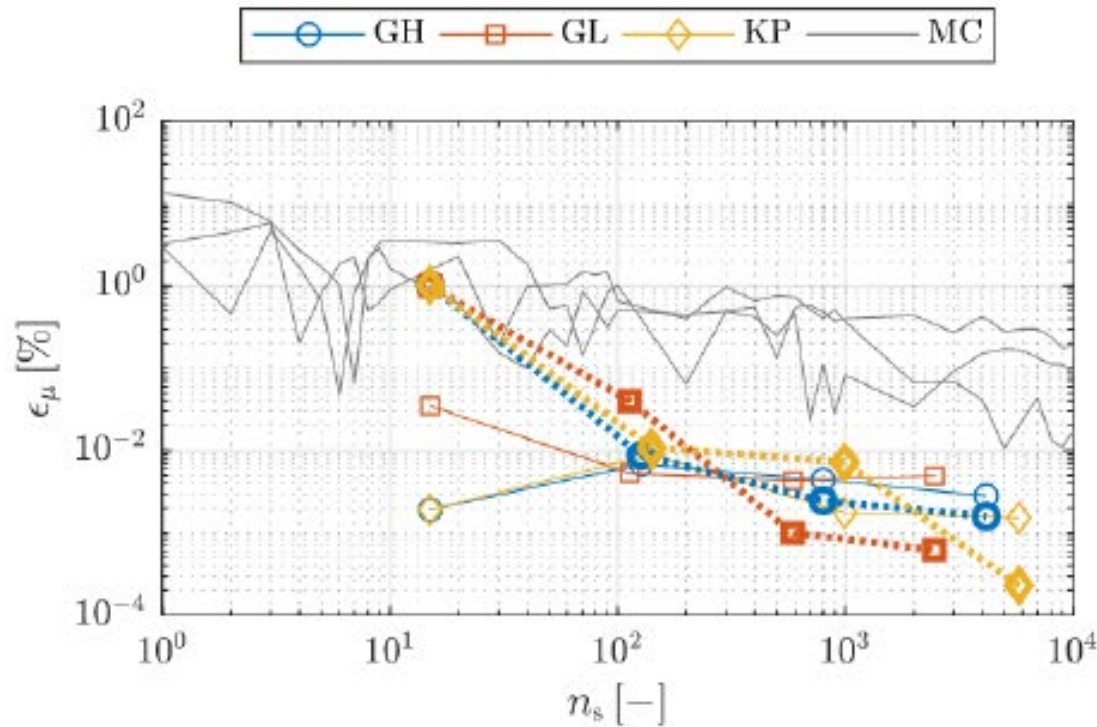


deterministic material parameters

$$E = 210 \text{ GPa}$$

$$\nu = 0.3$$

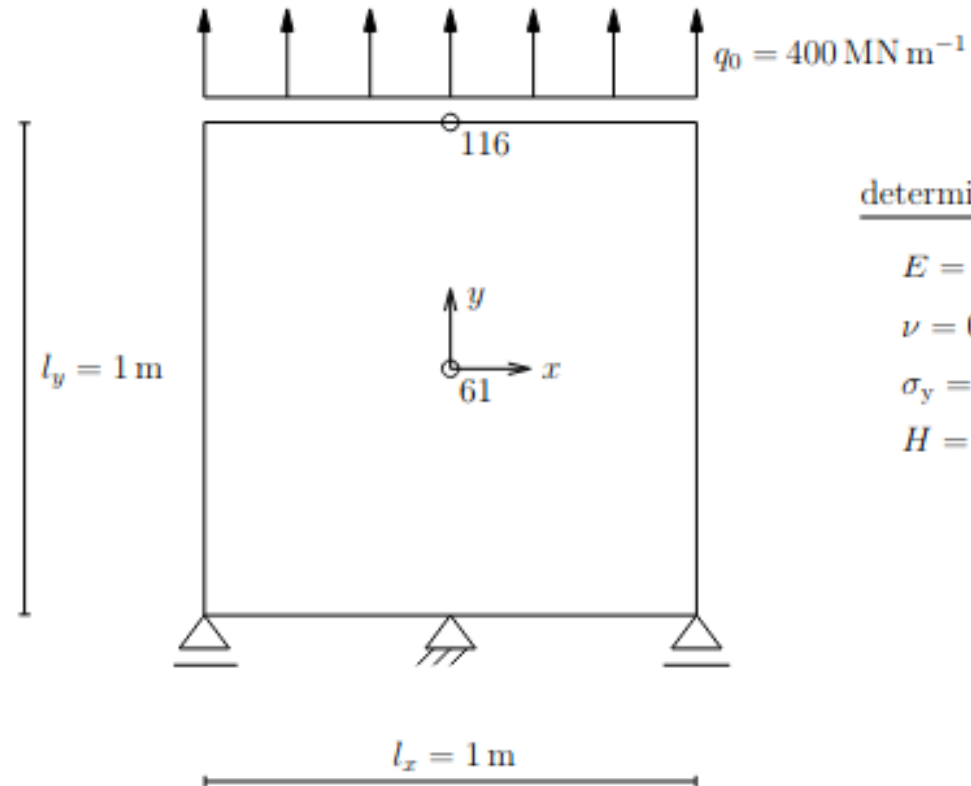
Numerical Example



Way faster convergence
compared to MC for
sufficient smooth problem.

Numerical Example

Elasto-plastic problem with
Young's modulus and yield
stress as random fields



deterministic material parameters

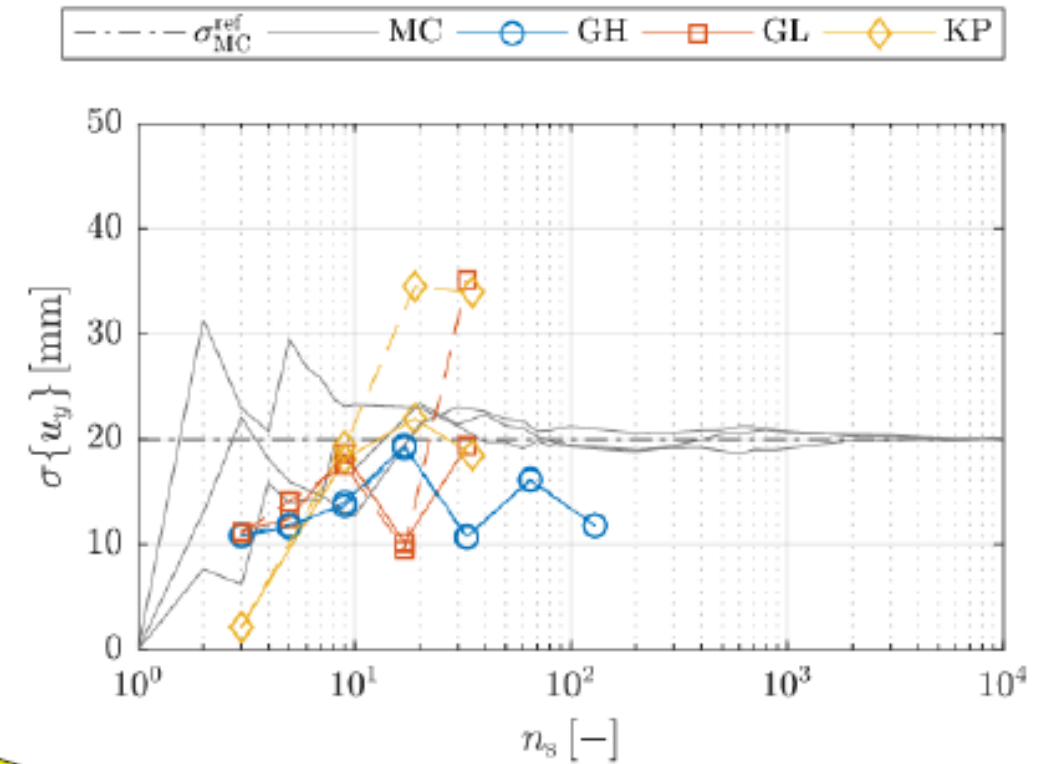
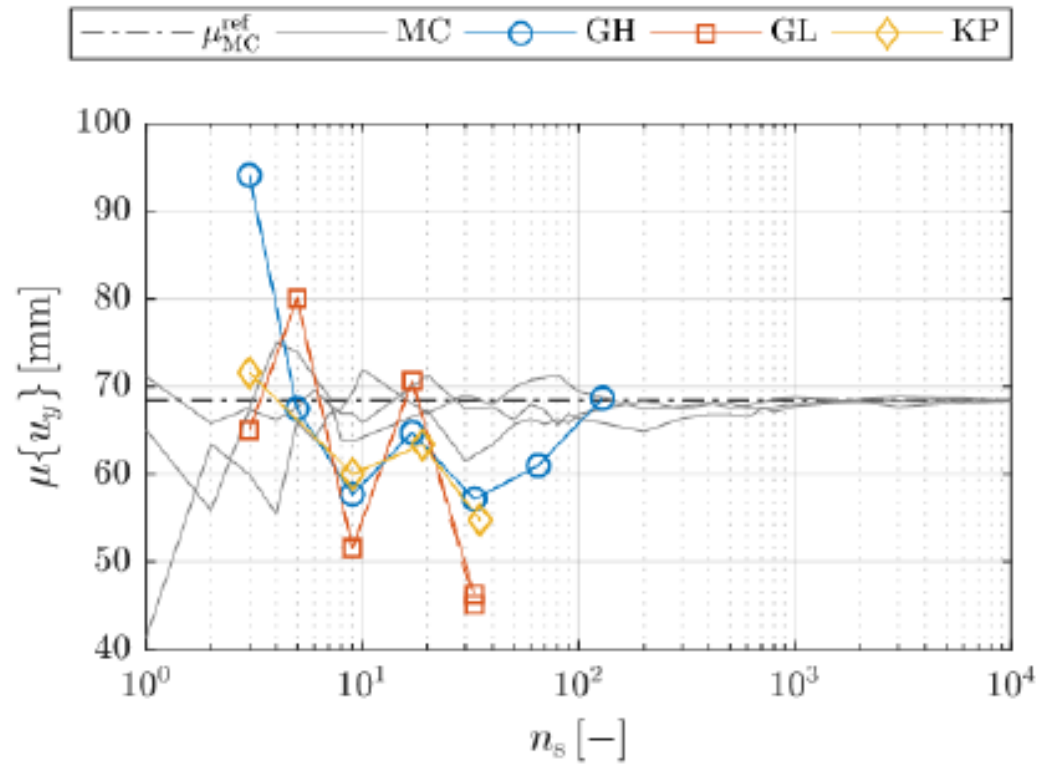
$$E = 210 \text{ GPa}$$

$$\nu = 0.3$$

$$\sigma_y = 300 \text{ MPa}$$

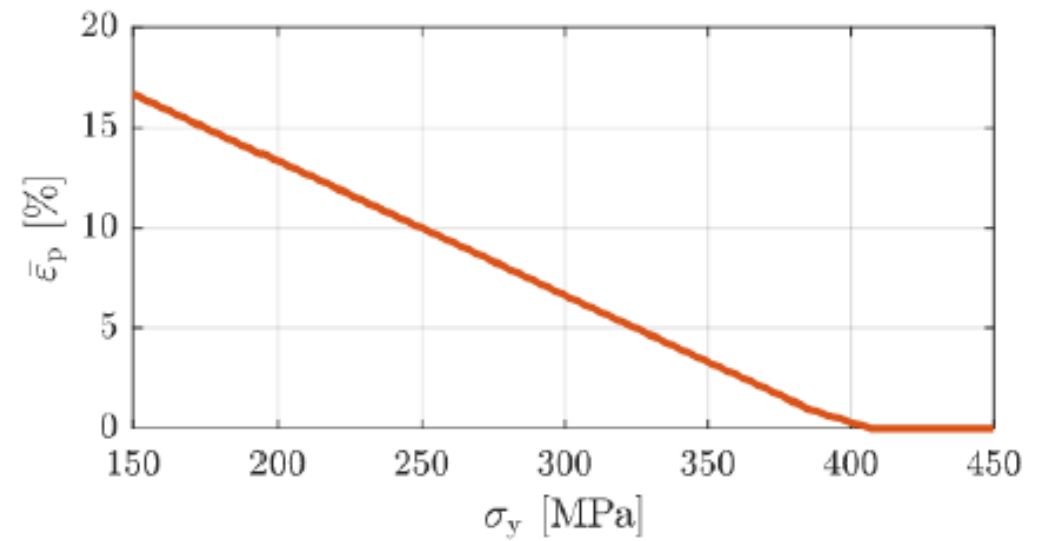
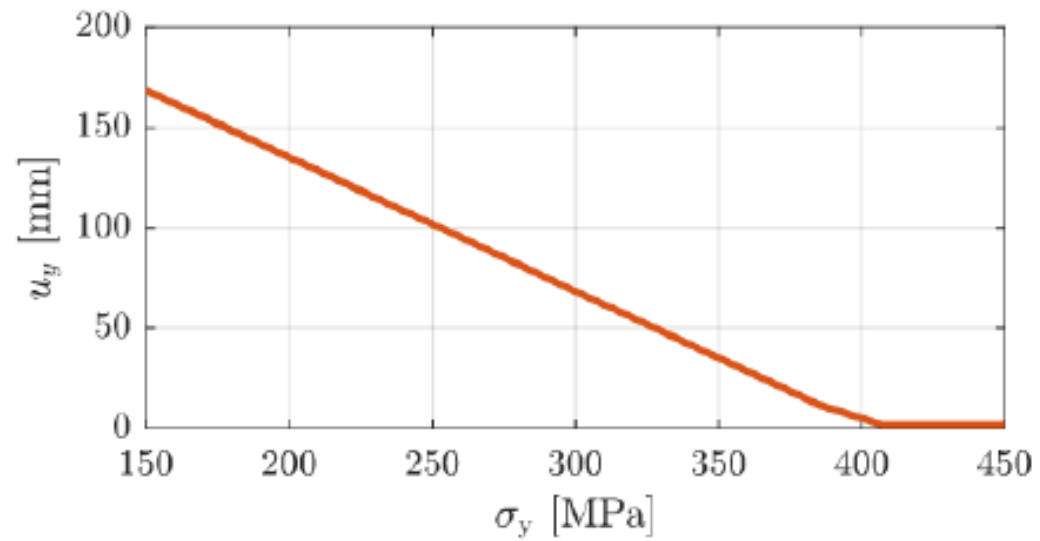
$$H = 1.5 \text{ GPa}$$

Numerical Example



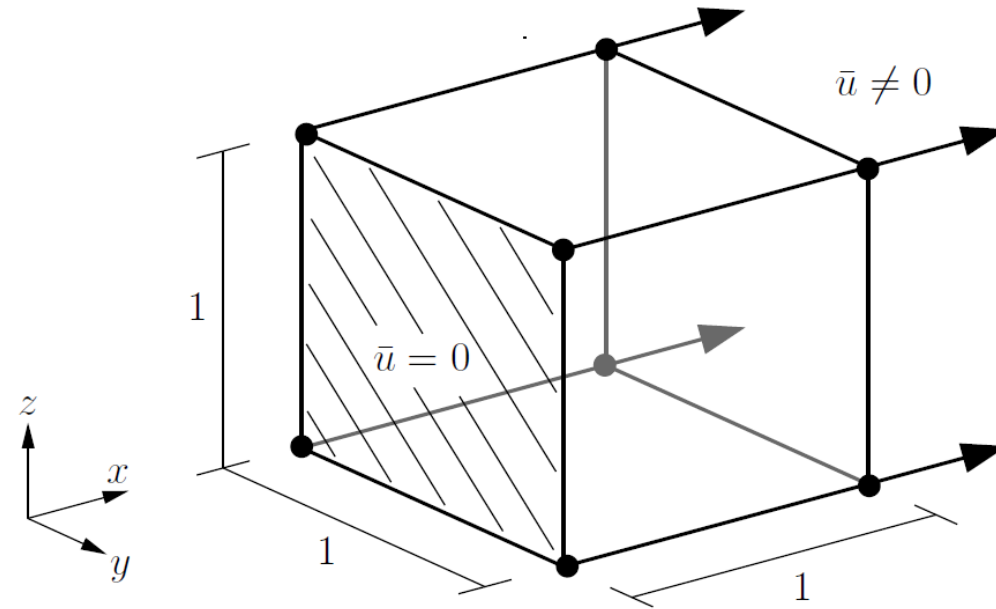
No convergence behavior!

Numerical Example



Non-smooth-response.

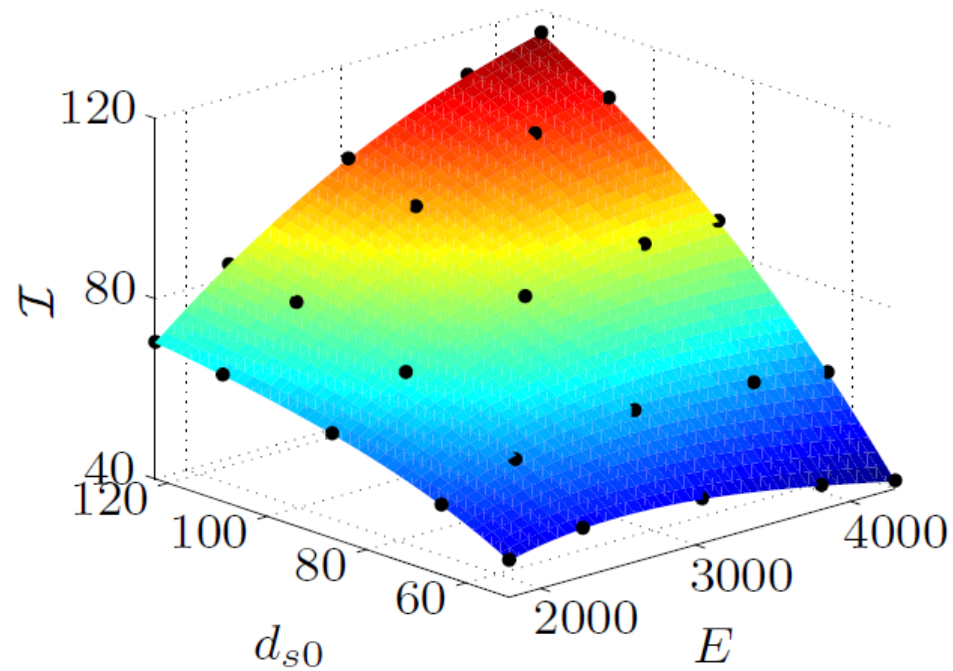
Brittle Damage model:



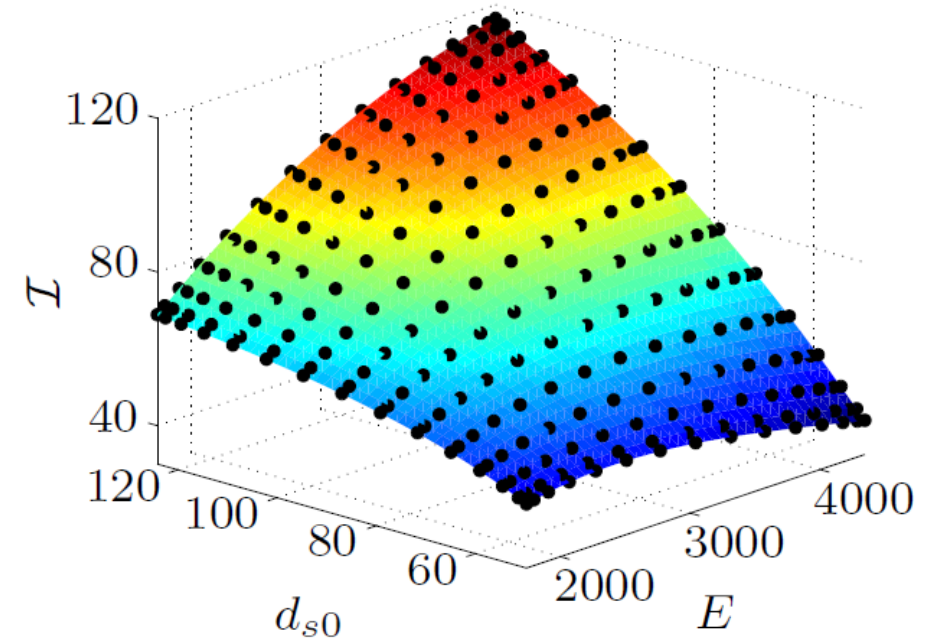
Parameter	Verteilung	$\langle \bullet \rangle$	SD _% ($\bullet$) [%]
E	lognormal	2800 N/mm ²	[2, 5, 10, 20]
d_{s0}	lognormal	80	[2, 5, 10, 20]
ν	-	0.3	-
D_{∞}	-	0.99	-

Numerical Example: Response Surfaces

Full Tensor Product



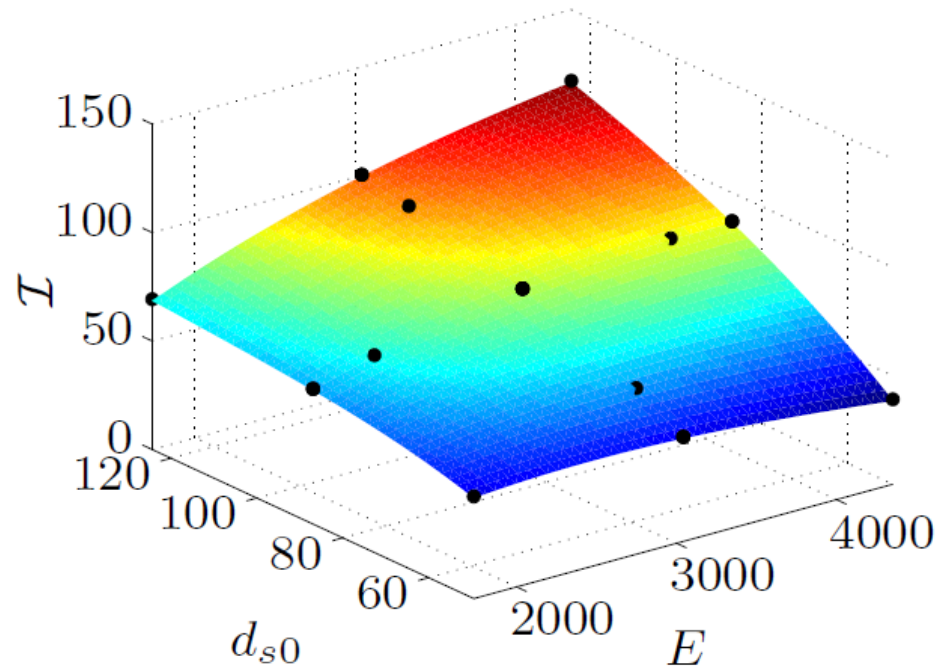
$SD = 20\%$, level $k = 5$.



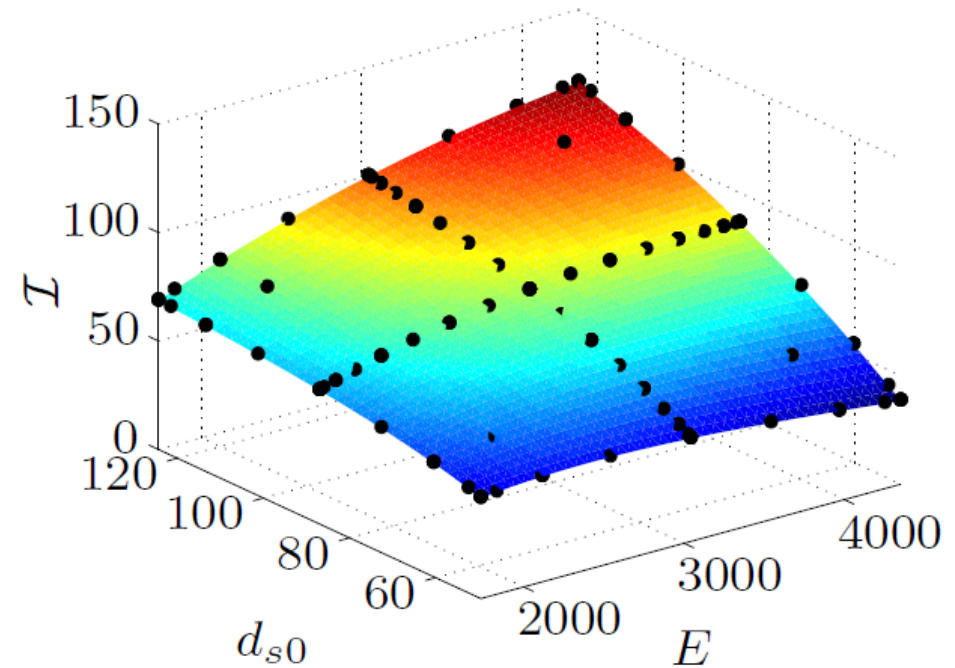
$SD = 20\%$, level $k = 15$.

Numerical Example: Response Surfaces

Smolyak method



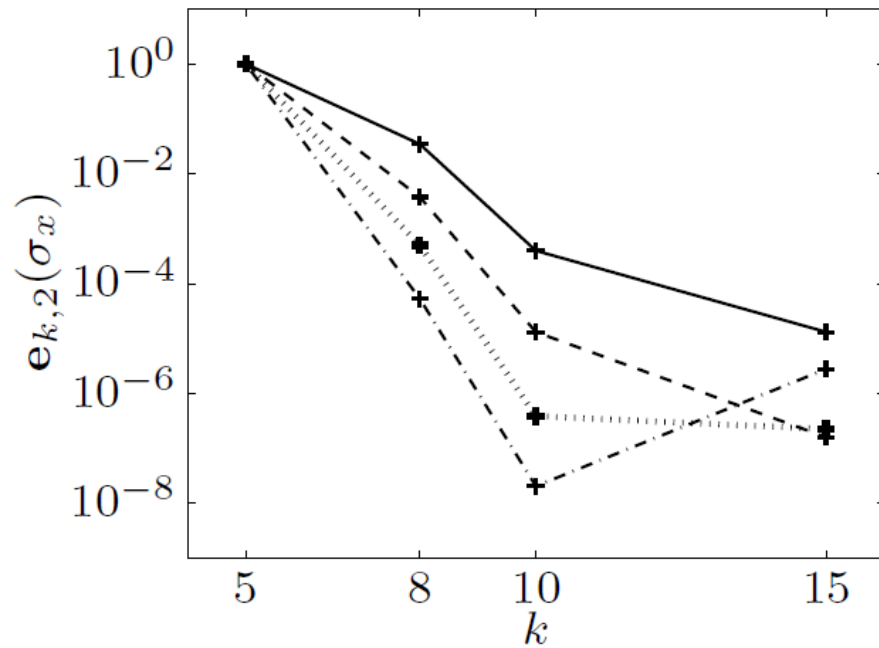
$SD = 20\%$, level $k = 5$.



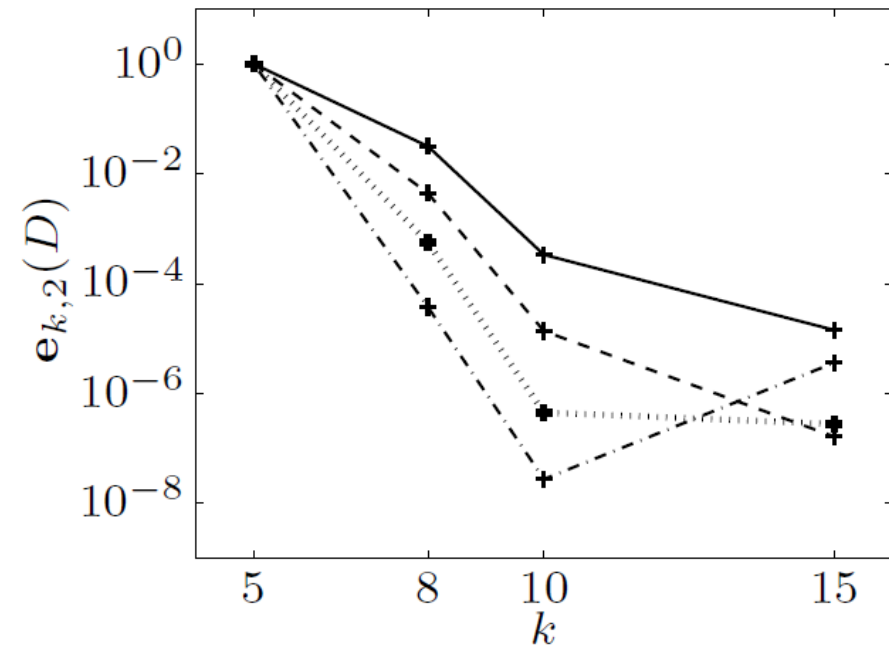
$SD = 20\%$, level $k = 15$.

Full Tensor Product

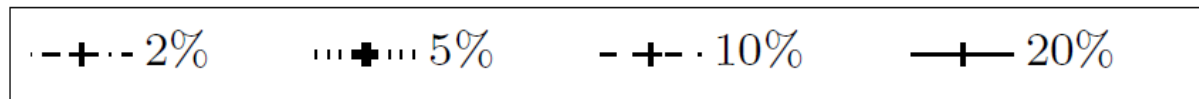
$$\mathbf{e}_{k,N}(\bullet) = \frac{\|\mathcal{I}(k-1, N)(\bullet) - \mathcal{I}(k, N)(\bullet)\|}{\mathcal{I}(k-1, N)(\bullet)}, \quad \text{für } k \geq 2$$



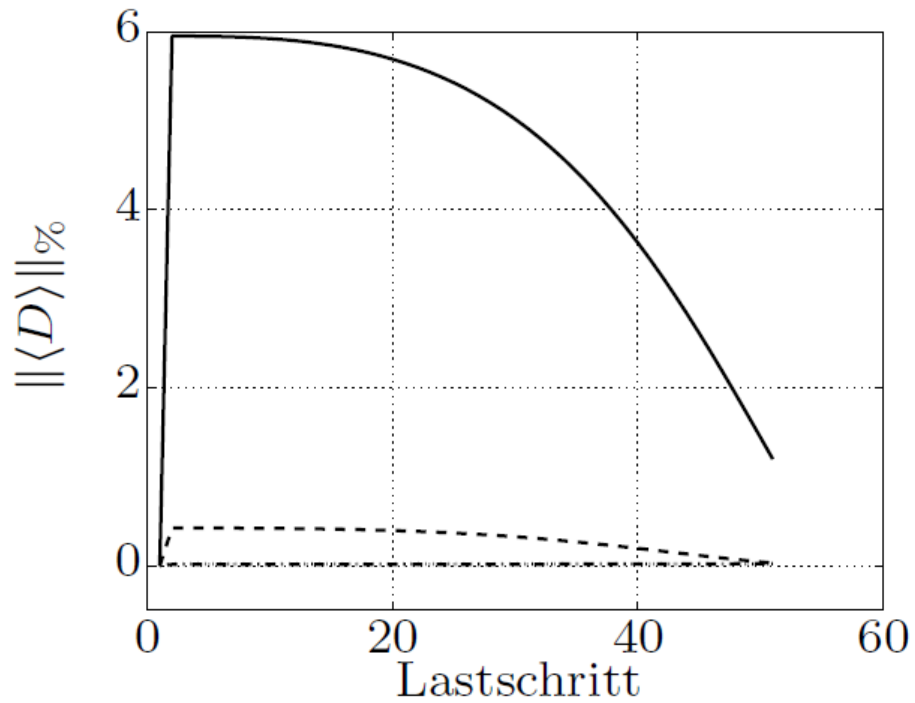
(a) Spannung σ_x



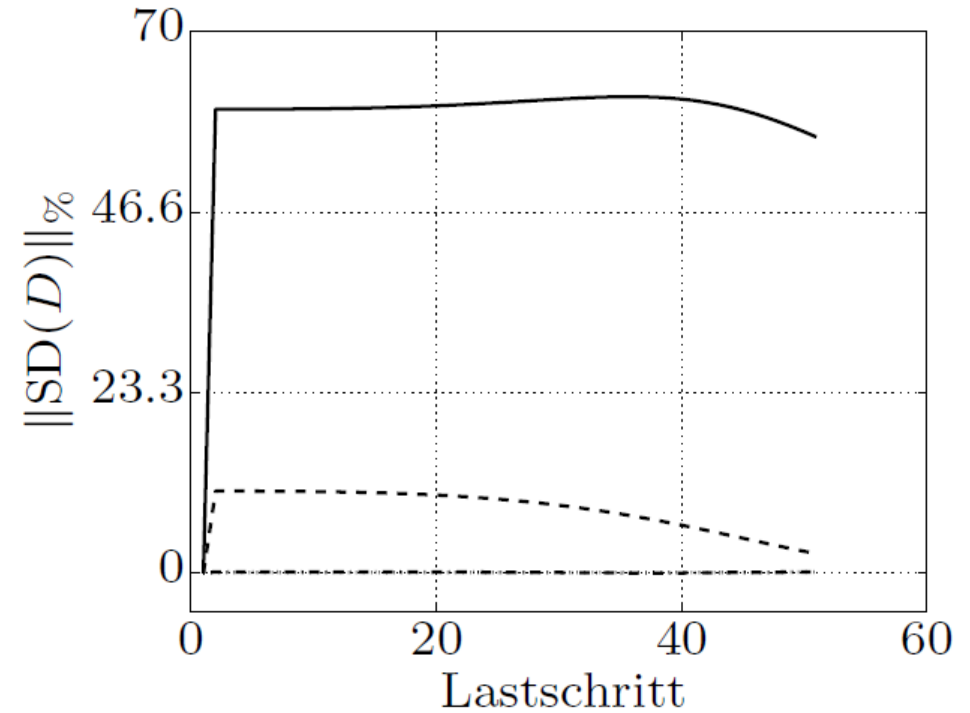
(b) Schaden D



Deviation from MCS



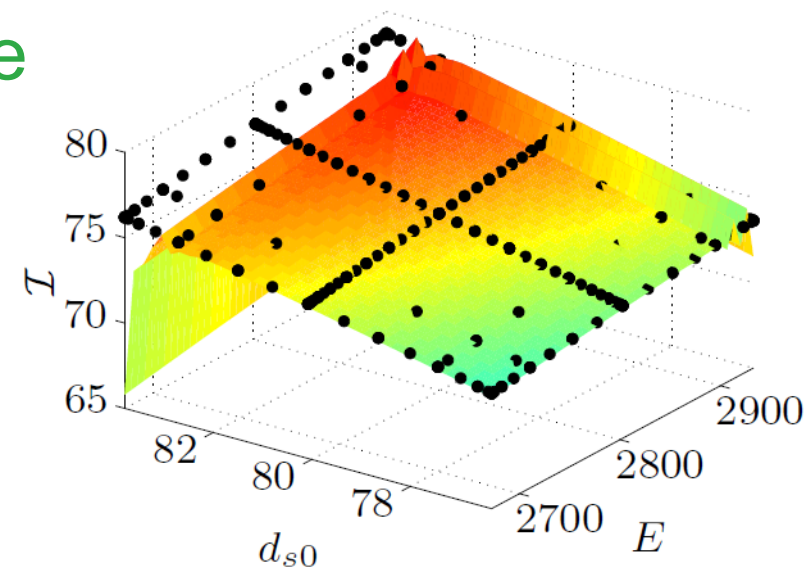
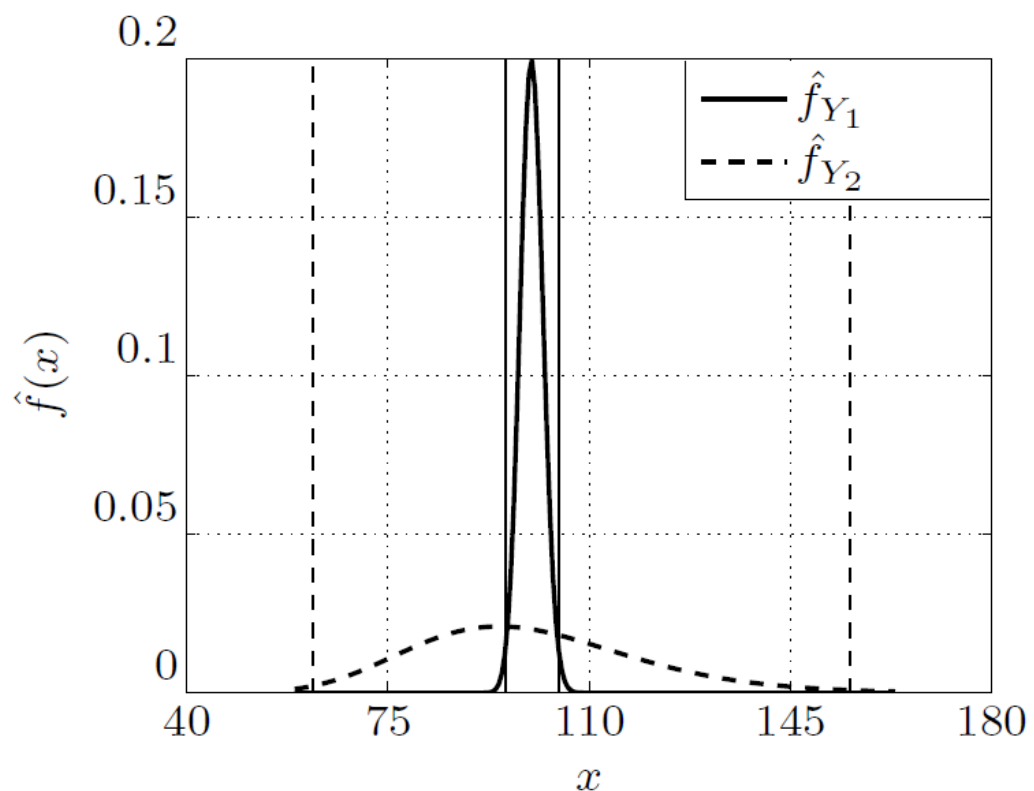
(b) Erwartungswert $\langle D \rangle$



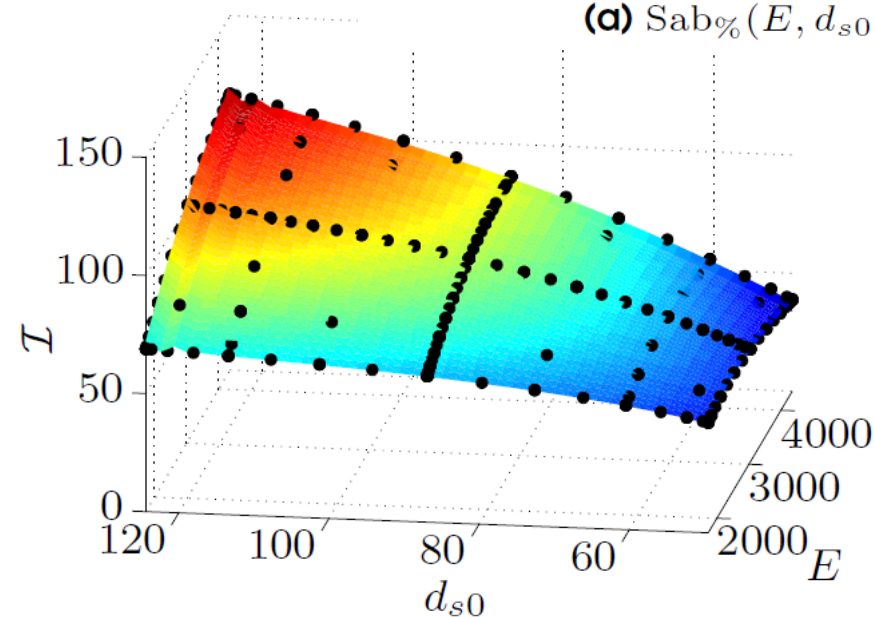
(d) Standardabweichung $SD(D)$

..... $k = 10$ -.-.- $k = 8$ - - - $k = 5$ — $k = 3$

On the Smoothness of the Response

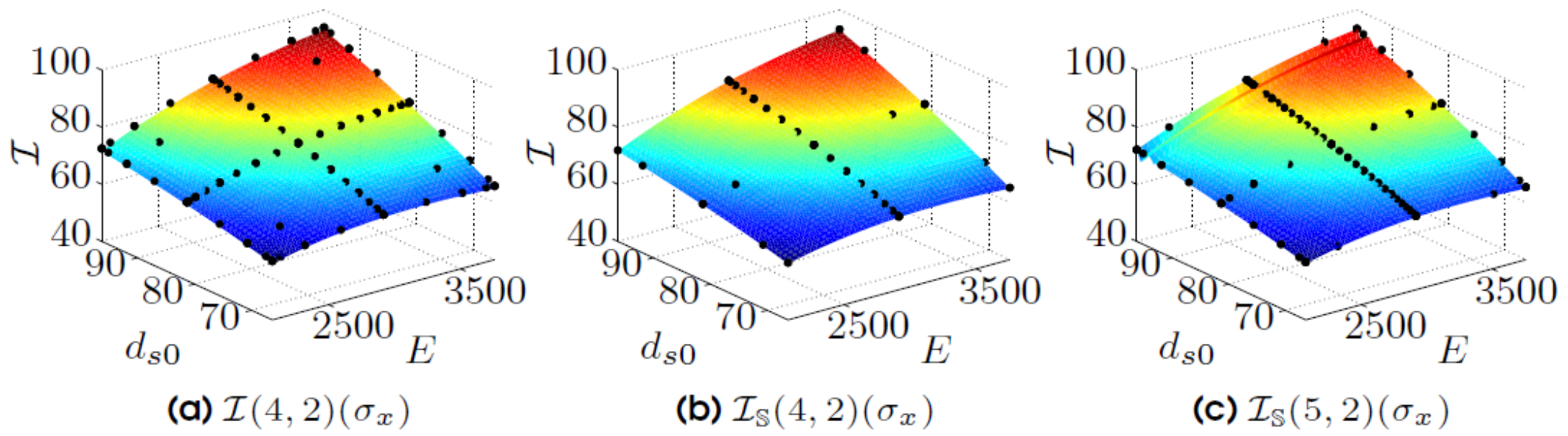


(a) $\text{Sab}\%(E, d_{s0}) = 2\%$



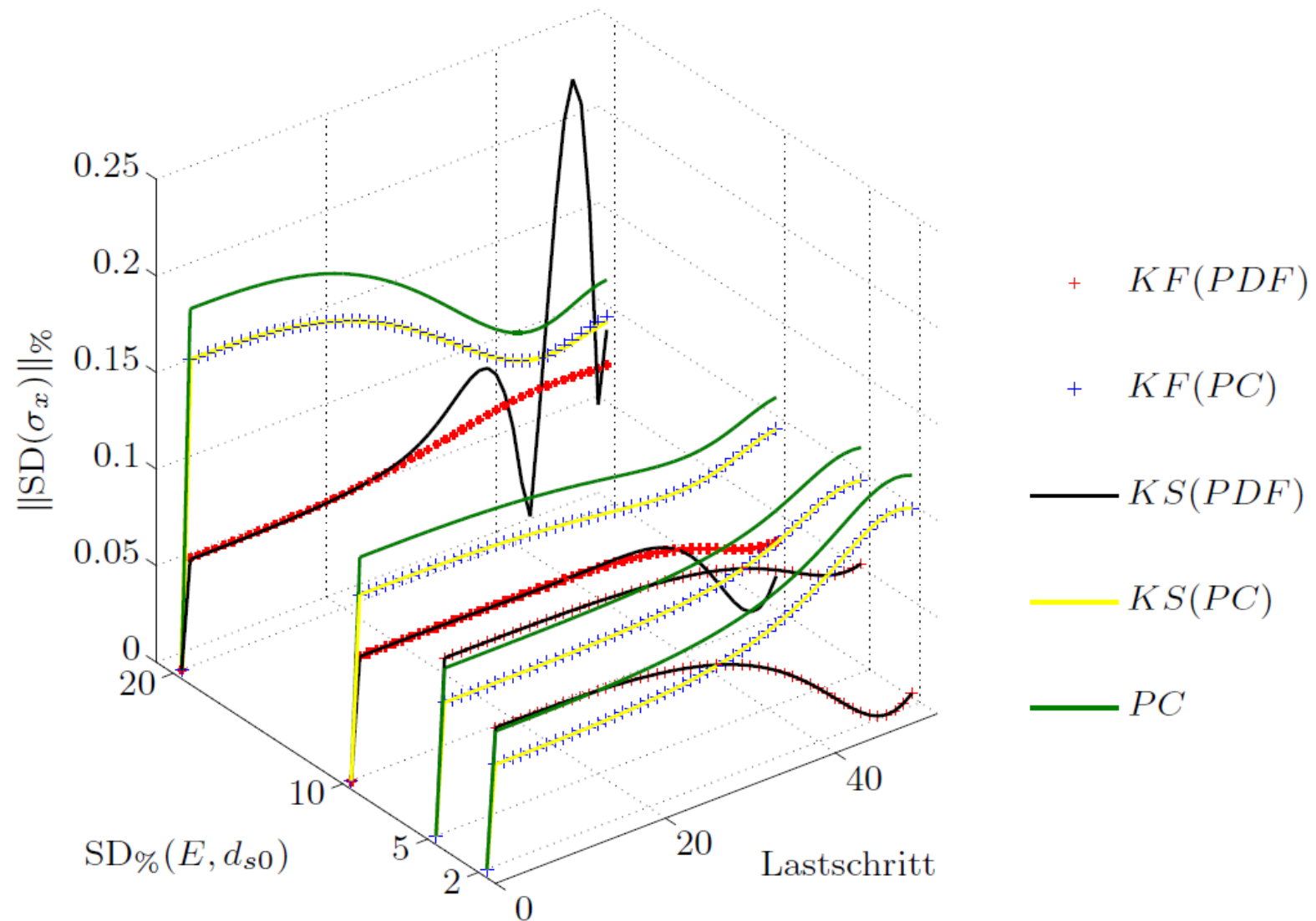
(b) $\text{Sab}\%(E, d_{s0}) = 20\%$

Anisotropic Smolyak



$SD_{\%}(d_{s0})$ [%]	$SD_{\%}(E)$ [%]	α	$\#\mathcal{H}_s$ (CC)	$\#\mathcal{H}_s$ (GH)
5	5	$[1 \ 1]$	65	131
	6	$[1.2 \ 1]$	37	67
	7	$[1.4 \ 1]$	33	58
	8	$[1.6 \ 1]$	29	50
	9	$[1.8 \ 1]$	29	50
	10	$[2 \ 1]$	29	50

Comparison of Different Approaches



Summary

- » Stochastic collocation methods are sampling based, FE-Solver called at certain points in the random space (collocation grid points)
- » Polynomial fit in the multi-dimensional stochastic space (response surface)
- » Computational effort and accuracy depends on the choice of the collocation points
- » Fast convergence for sufficient smooth problems
- » Advanced methods (multi-element, adaptive, ...) to tackle problems with non-smoothness, discontinuities

Alternatives to Lagrange Interpolation

- piecewise (adaptive) grids, e.g. hat-functions.
- Polynomial Chaos
- Element wise Polynomial Chaos
- Sparse adaptive PC

Polynomial Chaos Expansion

Generation of a Response Surface with Chaos Polynomials based in sampling,

$$n_{PC} \ll n_{MC}$$

Model approximation

$$\tilde{Y}(\boldsymbol{x}, \hat{\boldsymbol{z}}) = \tilde{\mathcal{M}}(\boldsymbol{x}, \hat{\boldsymbol{z}}) = \sum_{\boldsymbol{\alpha} \in \mathcal{A}} \hat{a}_{\boldsymbol{\alpha}}(\hat{\boldsymbol{z}}) \Psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$$

Statistical evaluation

$$\begin{aligned} \mu\{\tilde{Y}\}(\hat{\boldsymbol{z}}) &= \hat{a}_0(\hat{\boldsymbol{z}}), \\ \sigma^2\{\tilde{Y}\}(\hat{\boldsymbol{z}}) &= \sum_{\boldsymbol{\alpha} \in \mathcal{A} \setminus 0} \hat{a}_{\boldsymbol{\alpha}}^2(\hat{\boldsymbol{z}}). \end{aligned}$$

Polynomial Chaos Expansion

Truncation schemes

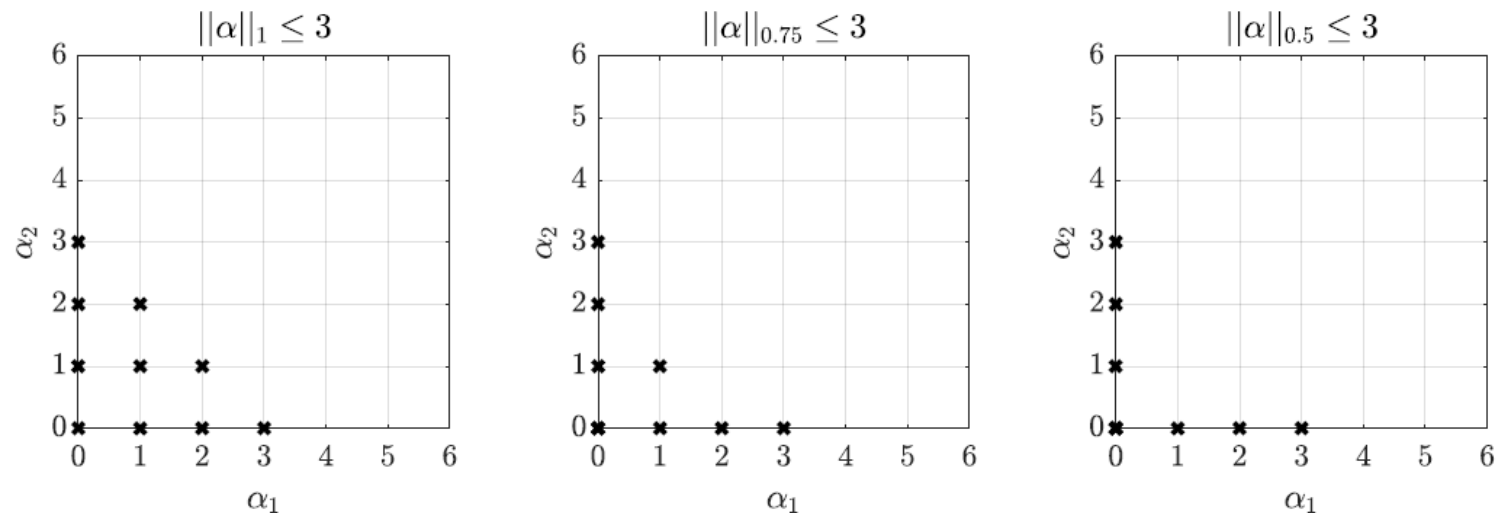
$$\mathcal{A}(N, p) = \{\alpha \in \mathbb{N}^N : |\alpha| \leq p\}$$

$$|\alpha| = \alpha_1 + \dots + \alpha_N.$$

Hyperbolic truncation of the polynomial degree p

$$\mathcal{A}(N, p, q) = \{\alpha \in \mathbb{N}^N : \|\alpha\|_q \leq p\}$$

$$\|\alpha\|_q = (\alpha_1^q + \dots + \alpha_N^q)^{1/q}.$$



(a) $p = 3$